

ECO 102 Midterm Practice Questions

Q1) Consider an economy that produces and consumes DVDs and Cars. The following data are given for two different years.

	Year 2000	Year 2010
Price of a Car	\$50,000	\$60,000
Price of a DVD	\$10	\$20
Number of cars produced	100	120
Number of DVDs produced	500,000	600,000

- (a) Using 2000 as the base year, calculate the Nominal GDP and the Real GDP.
- (b) Calculate the GDP Deflator for 2010.
- c) Calculate the Economic Growth Rate.

Q2) What effect would a decrease in the price level have on the wealth of the people in the economy and how would it affect the quantity of real GDP demanded?

Q3) The following table reflects the purchases of an average consumer in Bangladesh.
(Assuming these consumers purchase only hand sanitizers and masks.) **Suppose 2018 is the base period.** Use the table to answer the following questions:

Goods consumed Year	Hand sanitizers		Masks	
	Price (taka per unit)	Quantity	Price	Quantity
2017	30	100	10	50
2018	40	200	6	55
2019	80	500	30	100

- a) Calculate the RPI/CPI of Bangladesh for 2017 and 2019.
- b) Compute the rate of inflation/deflation between 2017 and 2019. Why might not the rate of inflation calculated using the CPI/RPI index be an accurate measure of the cost of living? Explain.
- c) According to Bangladesh Bank data the inflation rate in Bangladesh as of October 2020 is 6.44%, which is around 1% higher compared to that of October 2019. Illustrate and explain in your own words, what sort of impact such persistent inflation can have on the real GDP and employment of the country.

Q4) Study the table and calculate the following

(a) RPI for the years

(b) Inflation rate

RPI basket for base year (2017)	Item	Quantity	Price (Tk)	Total Cost
	Radios	7	350	2450
	Apples	5	120	600
	Watches	10	400	4000
RPI basket for next year (2018)				7050
	Radios	9	370	7*370 = 2590
	Apples	7	125	625
	Watches	15	435	4350
				7565

Q5) The total population of India is 182 million. The people aged between 16 years to retirement totaled 72 million. Among this, 32 million people are considered to be economically inactive and the rest are active. If 18 million people are still unemployed, figure out

- i. Unemployment rate
- ii. Employment rate
- iii. Economic Activity Rate

Q6) Brazil's real GDP was 1,360 trillion in 2009 and 1,434 trillion in 2010. Brazil's population was 191.5 million in 2009 and 193.3 million in 2010. Calculate

- a. The economic growth rate.
- b. The growth rate of real GDP per person.

Q7) Japan's real GDP was 525 trillion yen in 2009 and 535 trillion yen in 2010. Japan's population was 127.6 million in 2009 and 127.5 million in 2010. Calculate

- a. The economic growth rate.
- b. The growth rate of real GDP per person.

Q8) Bangladesh's government is running on a budget deficit. A recent announcement has informed the citizens that the government would be planning to cut down on its expenses. What type of policy is the Bangladeshi government planning to use? What changes can the new policy bring to the AD-AS model, if any? Show using a diagram.

Q9)

2017	Price	Qty
Books	18	20
Magazines	5	15
2018	Price	Qty
Books	20	30
Magazines	7	20

Holding 2017 as the base year, calculate

- a) Nominal GDP and Real GDP in 2017
- b) Nominal GDP and Real GDP in 2018
- c) Economic Growth Rate from 2017 to 2018

Q10) Economists and businesses believe that the second wave of the Covid-19 pandemic would affect the country's economic pillars-export, investment and private consumption-severely, as the tenure of the wave is likely to be longer. Some of the major trading partners of Bangladesh (mainly the European countries) have reintroduced either lockdowns or equivalent steps amid rising virus infections and deaths. These countries have already squeezed overall supply and demand due to the impact of the second wave of the pandemic, resulting in an adverse impact on Bangladesh's export and investment.

The second wave has already reduced the export orders of the readymade garment sector by 30%. Zahid Hussain, former lead economist of the World Bank Dhaka office recently commented "Export, investment and private consumption sectors *will ... be affected* severely very soon as the consumers' confidence is gradually reducing."

- a) With regard to the above excerpt, how do you think the real wage rate and the real GDP of the Bangladesh economy will be affected? Explain and Illustrate your answer with an appropriate diagram.
- b) Based on the above, **how do you think the AD-AS model would change** AND mention the kind of output gap that the economy might experience.

Q11) The pandemic has led to a sharp decline in most markets. Regardless, certain economies have experienced rising price levels during the lockdown. Explain what type of inflation these economies are experiencing and how they have occurred.

Q12) Suppose a country has recently experienced a technological breakthrough. Briefly discuss the effects of this on the labor market, productivity and output.

Solutions

1. a. For 2010,

$$\text{Real GDP} = P (\text{base year}) * Q (\text{current year})$$

$$= (50,000 * 120) + (10 * 600,000)$$

$$= 6,000,000 + 6,000,000 = 12,000,000$$

$$b. = (\text{Nominal GDP} / \text{Real GDP}) * 100 = 160\%$$

$$c. \text{Econ. Growth Rate} = [\text{Real GDP (current year)} - \text{Real GDP (last year)}] / \text{Real GDP (last year)} * 100$$

$$(12,000,000 - 10,000,000) / (10,000,000) * 100 = 20\%$$

2. A decrease in Price would increase the value of wealth and hence, when one has more money, it can buy more. So, the qty of RGDP demanded increases.

3. a. We know, $RPI = \frac{\text{Cost of basket at current period price}}{\text{Cost of basket at base period prices}} * 100$

Cost of basket at base period prices

Here,

$$\text{Cost of basket in 2017: } (30 \times 200) + (10 \times 55) = \text{BDT } 6550$$

$$\text{Cost of basket in 2018: } (40 \times 200) + (6 \times 55) = \text{BDT } 8330$$

$$\text{Cost of basket in 2019: } (80 \times 200) + (30 \times 55) = \text{BDT } 17650$$

Given that 2018 is the base-period

RPI 2017: $\frac{6550}{8330} \times 100$

= 78.63

RPI 2019: $\frac{17650}{8330} \times 100$

= 211.88

b. We know, Inflation Rate = $\frac{(\text{RPI this year} - \text{RPI last year})}{\text{RPI last year}} \times 100$

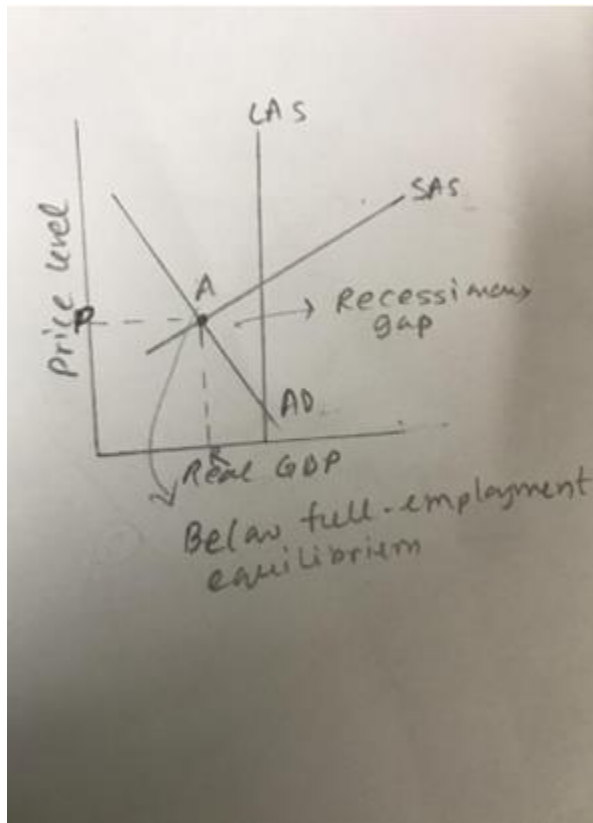
Here the inflation/deflation rate between 2017 and 2019 is to be determined so,

Inflation Rate: $\frac{(211.88 - 78.63)}{78.63} \times 100$

= 169.46

The RPI is not a perfect measure of the price level, changes in the RPI can overstate the inflation rate. This is because of the biases of the price index. Here in this case the inflation might be biased because the quality of the masks and sanitizer is undergoing quality improvement with every coming day for more protection against the virus. Such Kn95 masks have been more in demand than surgical masks and there are more effective sanitizers in the market than before. These kinds of changes increase the price, but such price increases are not

inflation. Again , there is substitute bias that hand sanitizer is competing with the soap so if substitution is somewhat cheaper then consumers make substitutions , so price index based on a fixed basket overstates the effects of a given price change on the inflation rate. So it can be said that we might not get a clear view of the cost of living from the rate of inflation calculated from RPI/CPI .



c. As we know the persistent inflation leads to falling real income as a result economic growth is being hampered. And this is because of the instability and consumption slump by the pandemic and its lockdown .The GDP has been lowered and also unemployment in the economy. Due to the rise in price level of the goods and services the AD falls short compared to the As as seen in the figure. So that means the willingness of the consumers to buy goods in the country falls. Thus the quantity of the real GDP demanded falls. With the demands for goods and services falling the workers are not employed rather are laid off their jobs thus causing unemployment even. Therefore a below employment point is denoted as A. Thus the potential GDP of the country is now higher than real GDP as the resources of the country and the laborers are not utilized to its full potential. This output gap is known as recessionary gap.

4. a. 2017

RPI = (Cost of basket at the current period prices) / (Cost of the basket at the base period prices) * 100

$$= 7050 / 7050 * 100 = 100\%$$

2018

$$RPI = 7565 / 7050 * 100 = 107.3 \%$$

b) We know, Inflation Rate = $\frac{(RPI \text{ this year} - RPI \text{ last year})}{RPI \text{ last year}} \times 100$

RPI last year

Here the inflation/deflation rate between 2017 and 2018 is to be determined so,

Inflation Rate: $\frac{(107.3-100)}{100} \times 100$

100

$$= 7.3$$

5. i. Economically active / Workforce = 72-32 = 40

Unemployment rate = No of people unemployed / workforce * 100

$$= 18 / 40 * 100 = 45\%$$

ii. No of people employed / working age population * 100

$$= 22 / 72 * 100 = 30.56\%$$

iii. (Workforce/ working age population) *100

$$= (40/72) * 100$$

$$= 55.56\%$$

6.a) 5.44%

b) 4.46%

7.a) 1.90%

b) 1.98 %

8) Contractionary Fiscal Policy

AD shifts to the left. P falls. $RGDP < PGDP$.

As firms' production becomes cheaper, the firms would want to produce more. SAS would shift to the right. $RGDP = PDGP$

9.a) $2017 \text{ NGDP} = (18 \times 20) + (5 \times 15) = 360 + 75 = 435$

$$RGDP = NGDP \text{ in base yr} = 435$$

b) $2018 \text{ NGDP} = (20 \times 30) + (7 \times 20) = 740$

$$RGDP = (18 \times 30) + (5 \times 20) = 640$$

c) $GDP \text{ growth rate} = (640 - 435) / 435 * 100$

$$= 205 / 435 * 100$$

$$= 0.47 * 100$$

= 47.13%

10. a)

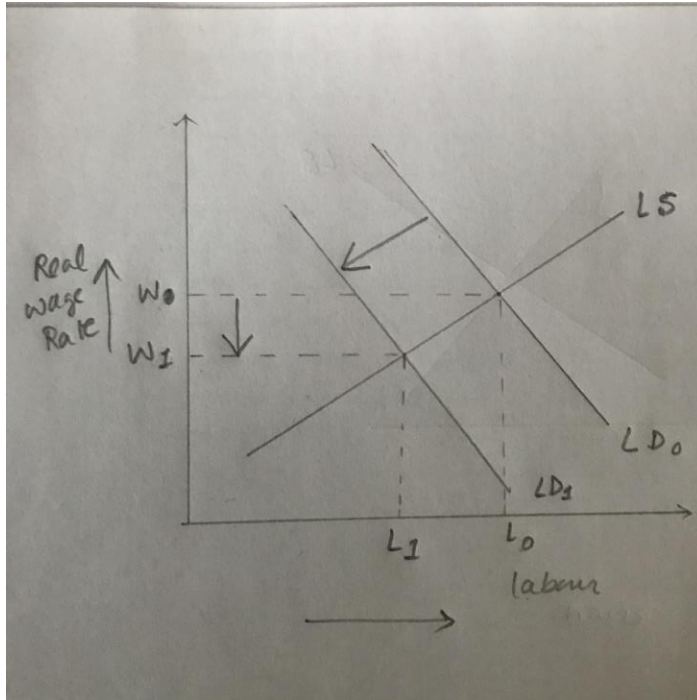


Fig 1

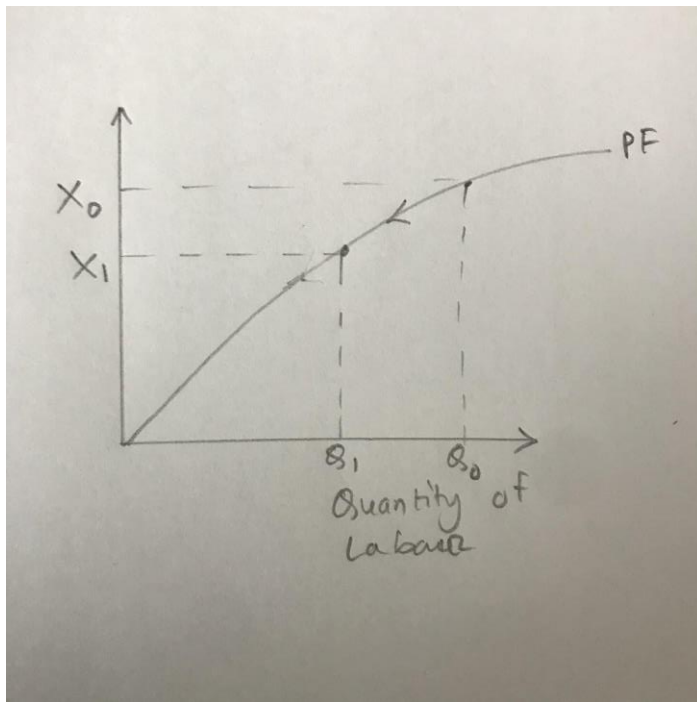
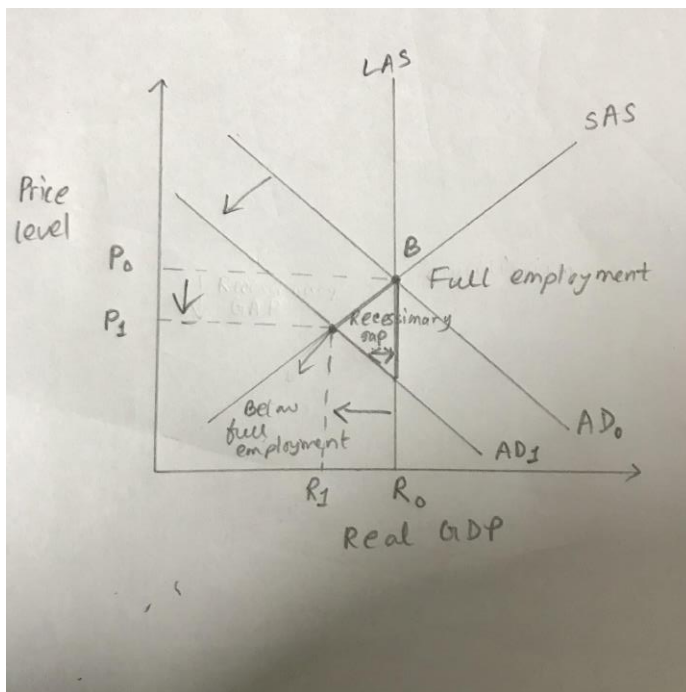


Fig 2

As from the figure 1 and 2 , we can see the labour demand curve shifted to the left. This is due to the second wave of the pandemic, lockdown is reintroduced. It has already reduced the

export orders of the readymade garments sector by 30 %. It would bring an impact in the labor market which will result in loss of jobs. The figure shows a decrease in the real wage rate from W_0 to W_1 with the decrease in the Labour demand, that is the number of labor from the L_{D0} to L_{D1} . The decrease in real wage rate will then result in the decrease in the disposable income and the laborers earning less will consume less and therefore the Real GDP will therefore decrease too. Here when the lockdown is reintroduced the orders in the garments of our country will be canceled and as a result the industries will not have money to pay the laborers and they will cut of labor's wage as result the labor demand will decrease as well as the wage rate and simultaneously decrease the disposable income .Thus, the Real GDP will decrease too.

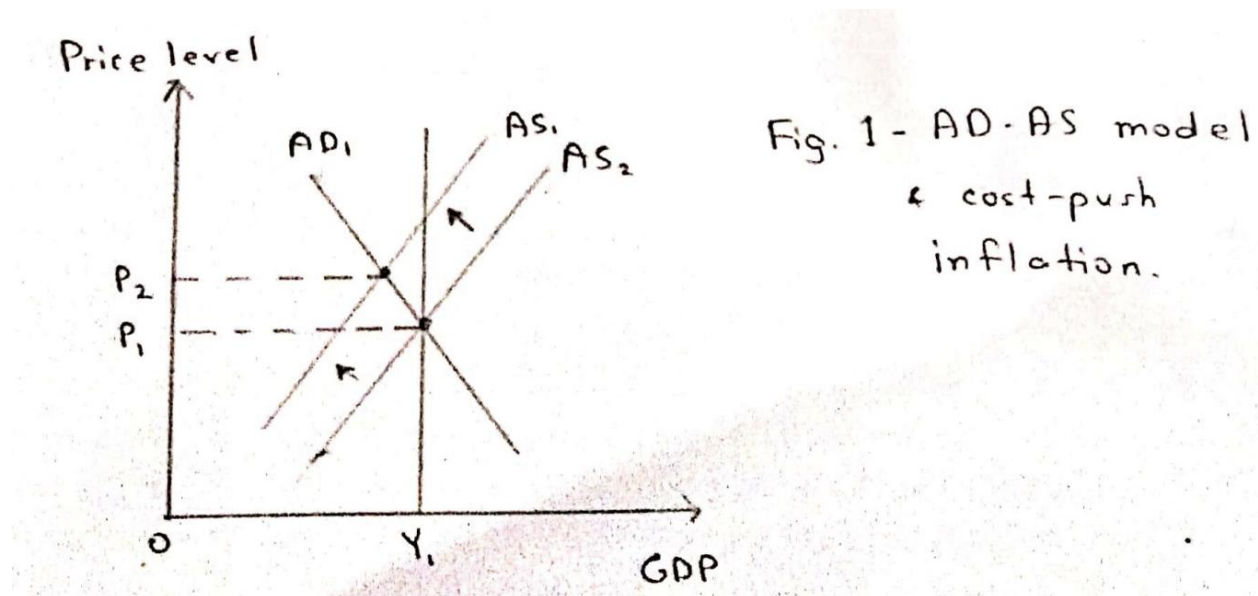
b.



From the figure of the AS-AD model it is seen that the AD shifted to the left side. When orders got canceled due to the lockdown of the second wave of the virus the orders got canceled out and thus it had an impact on the labor market for which jobs were lost and loss of jobs refer to loss of income and thus the fall of aggregate demand . Here the AD curve shift is shown in respect with the long run supply and the short run supply. The triangle formed in between the LAS SAS and AD_0 to AD_1 is the Recessionary gap. The recessionary gap is pointed on the

figure and this means there is recession because of the decrease of the real GDP. Thus the output gap here is the recessionary gap.

11) While there was a drop in demand in most markets due to the pandemic, there may also have been a fall in supply in most economies. This may have occurred due to lockdown shutting down factories and industries thereby causing a fall in overall output. Due to the disruption of global supply chains, international trade as well as rising unemployment, there may have been an increasing fall in aggregate supply. As shown in fig 1, a leftward shift from AS_2 to AS_1 , a rise in the price level may have led to cost push inflation. Here, we assume that economies were at full-employment equilibrium. Price- level consequently rose from P_1 to P_2 . This rise in price level may have been aggravated by a persistent fall in aggregate supply over time.



12) A technological breakthrough may lead to an increase in labor productivity. This may occur due to a rise in physical capital such as machinery which in turn may increase the output per labour. Using new technology makes work more efficient and faster thus promoting worker motivation and leading to more productivity. As shown in fig 1, labour demand may rise due to higher productivity and cause LD_1 to shift to LD_2 . This in turn increases the amount of labour hours from L_1 to L_2 .

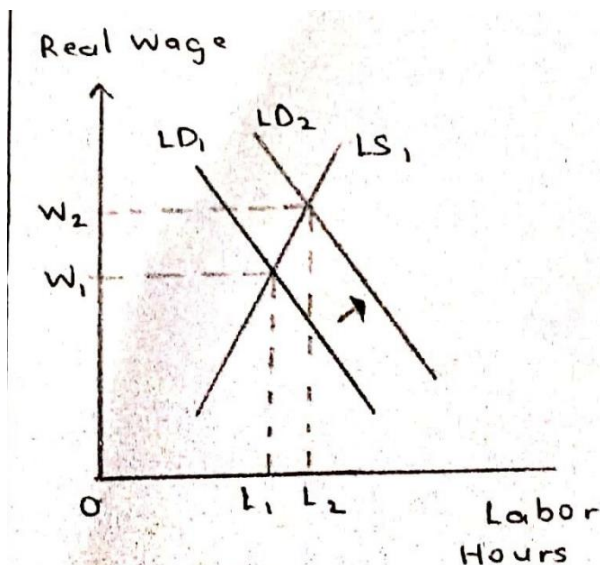


Fig. 1 - Labor Market

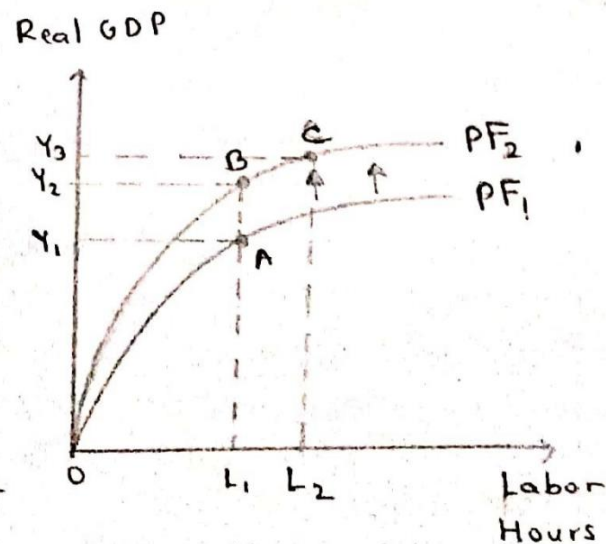


Fig. 2 - Aggregate Production Function

On the other hand, a rise in labour productivity leads to an upward shift in aggregate production from PF_1 to PF_2 as shown in fig 2. Thus, real GDP increases from A to B. As labour demand has increased, the rise in labour hours causes a movement along PF_2 from B to C. There is a further rise in output or real GDP from Y_2 to Y_3 . Overall, there is a rise in labour demand due to rising labour productivity followed by a rise in output and economic growth due to increasing potential GDP.

12) While there was a drop in demand in most markets due to the pandemic, there may also have been a fall in supply in most economies. This may have occurred due to lockdown shutting down factories and industries thereby causing a fall in overall output. Due to the disruption of global supply chains, international trade as well as rising unemployment, there may have been an increasing fall in aggregate supply. As shown in fig 1, a leftward shift from AS_2 to AS_1 , a rise in the price level may have led to cost push inflation. Here, we assume that economies were at full-employment equilibrium. Price- level consequently rose from P_1 to P_2 . This rise in price level may have been aggravated by a persistent fall in aggregate supply over time.

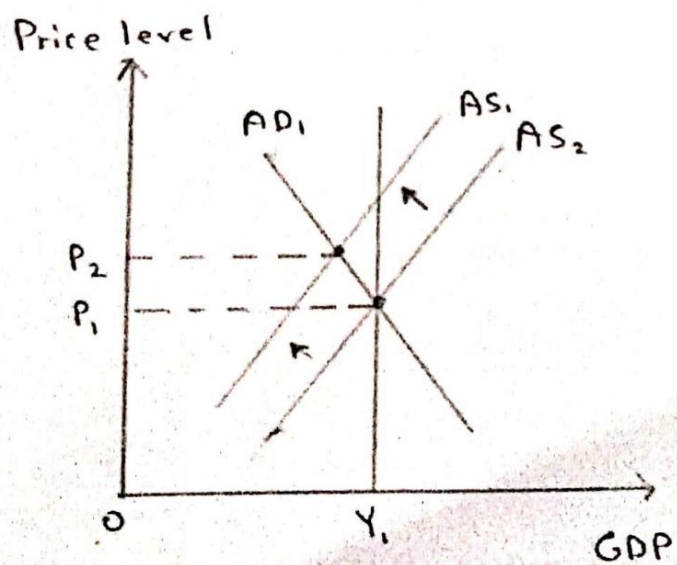


Fig. 1 - AD-AS model
& cost-push
inflation.